

Multi-scale Modeling of Hydrogen Transport in a Porous Fuel Cell Anode

Rosa Zhang^{1,2}, Anamika Chowdhury^{1,2}, Justin C. Bui^{1,2}, Clayton J. Radke¹, and Adam Z. Weber²

¹University of California, Berkeley, ²Lawrence Berkeley National Lab



Introduction: Hydrogen Transport Resistances

Gas transport in proton exchange membrane fuel cells encounters multiple length-scales and phases:

- 1) electrode pores at the length scale of electrode thickness (micrometer)
- 2) diffusion into the agglomerate through an ionomer film at the length scale of agglomerate radius (nanometer).

Herein, we present an analytical model of H₂ transport through porous anode catalyst layer (CL):

- **multi-scale:** 1+1D (along electrode thickness and catalyst particle radius)
- **multi-phase:** porous carbon agglomerate with Pt particles; ionomer film
- **purely analytical:** derived from spherical diffusion & Butler-Volmer kinetics

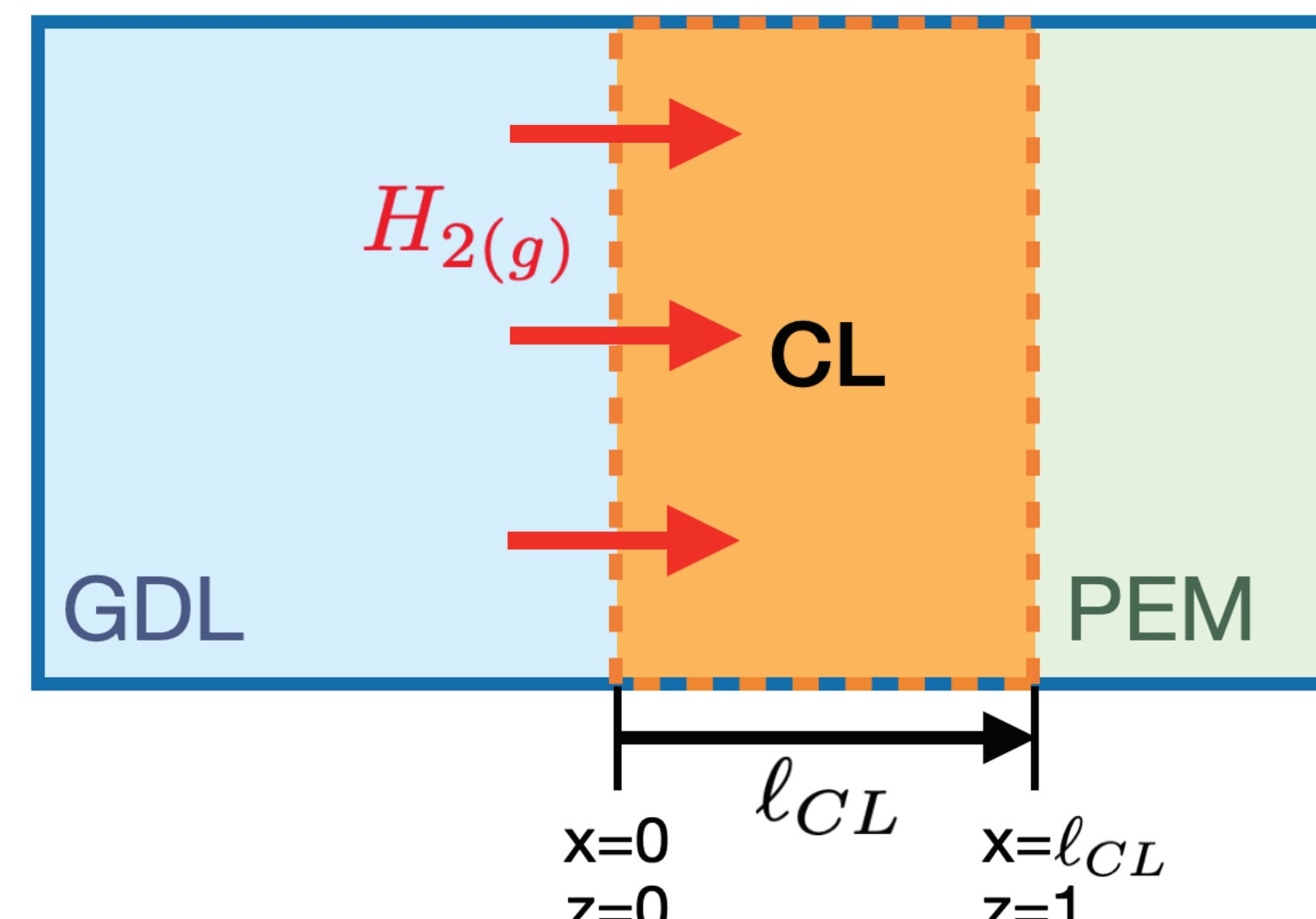


Figure 1. Diagram of 1-D micrometer-scale model of H₂ transport from the gas diffusion layer (GDL) through the porous electrode catalyst layer (CL), and onto the proton exchange membrane (PEM).

Multi-Scale Model: Derivation

I. Agglomerate

$$\frac{1}{z^2(R_a + \delta_f)} \frac{\partial}{\partial z} \left(z^2 D_{eff} \frac{\partial c_{H_2,agg}}{\partial z} \right) + r_{H_2} = 0$$

$$\text{BC 1: } \frac{\partial c_{H_2,agg}}{\partial z} \Big|_{z=0} = 0$$

$$\text{BC 2: } D_{eff} \frac{\partial c_{H_2,agg}}{\partial z} \Big|_{z=\frac{R_a}{R_a + \delta_f}} = D_{film} \frac{\partial c_{H_2,film}}{\partial z} \Big|_{z=\frac{R_a}{R_a + \delta_f}}$$

II. Ionomer film

$$\frac{1}{z^2(R_a + \delta_f)} \frac{\partial}{\partial z} \left(z^2 D_{eff} \frac{\partial c_{H_2,film}}{\partial z} \right) = 0$$

$$\text{BC 1: } c_{H_2,film} \Big|_{z=1} = S_{CL \rightarrow film} c_{H_2,CL}$$

$$\text{BC 2: } D_{eff} \frac{\partial c_{H_2,agg}}{\partial z} \Big|_{z=\frac{R_a}{R_a + \delta_f}} = D_{film} \frac{\partial c_{H_2,film}}{\partial z} \Big|_{z=\frac{R_a}{R_a + \delta_f}}$$

$$\text{BC 3: } c_{H_2,agg} \Big|_{z=\frac{R_a}{R_a + \delta_f}} = S_{film \rightarrow agg} c_{H_2,film} \Big|_{z=\frac{R_a}{R_a + \delta_f}}$$

III. Electrode thickness

$$D_{CL} \frac{\partial^2 c_{H_2}}{\partial x^2} + r_{H_2} = 0$$

$$\text{BC 1: } c_{H_2}(x=0) = c_{feed}$$

$$\text{BC 2: } \frac{\partial c_{H_2}}{\partial x} \Big|_{x=L} = 0$$

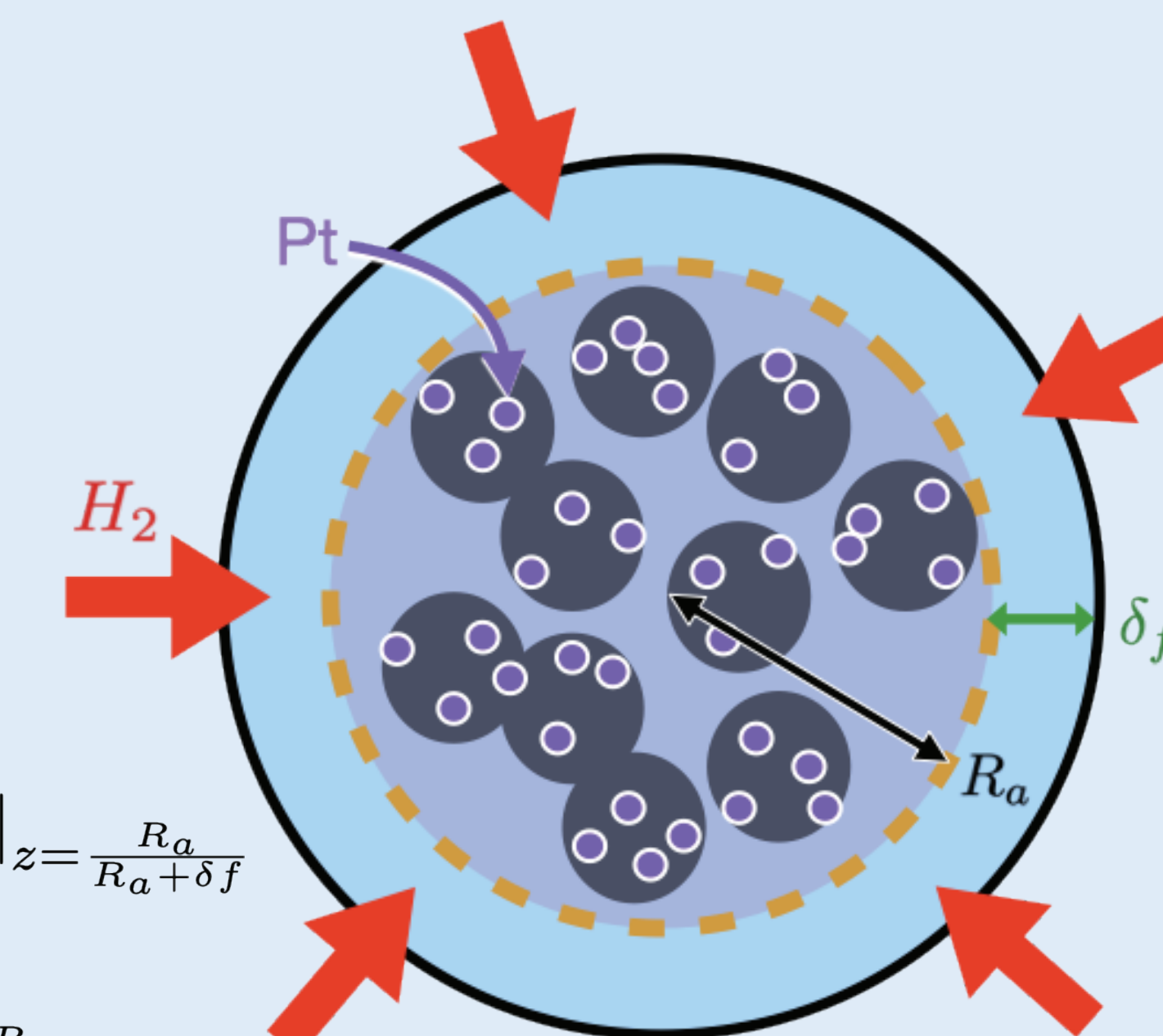


Figure 2. Diagram of radial H₂ transport into the catalyst particle through the ionomer film layer (δ_f) and into the agglomerate (R_a).

Derived analytical expressions

$$c_{H_2,agg} = \xi c_{H_2,CL} \frac{1}{\gamma} \frac{D_{film}}{D_{eff}} \sinh(\kappa z) \frac{1}{z}$$

$$c_{H_2,film} = c_{CL} S_{CL \rightarrow film} + \xi c_{H_2,CL} \left(1 - \frac{1}{z} \right)$$

$$c_{H_2,CL} = c_{feed} \left(\cosh(\sqrt{\nu} x) - \tanh(\sqrt{\nu} L) \sinh(\sqrt{\nu} x) \right)$$

I. Effectiveness Factor

$$E_R = \frac{W_{AR}}{W_{AR,0}} = \frac{\text{actual rxn rate in catalyst particle}}{\text{rxn rate assuming surface concentration}}$$

$$= \frac{D_{film} \xi}{\frac{1}{3} (R_a + \delta_f)^2 \left(\frac{a_v i_0}{2F} \right) \frac{1}{c_{ref}} \exp \left(\frac{\alpha_a F}{RT} (E - E_0) \right)}$$

II. Current

$$i = A \left(-\frac{3a_v i_0 (1 - \phi_c)}{c_{ref}} \right) \left(\exp \left(\frac{\alpha_a F}{RT} (E - E_0) \right) \right) \int_0^{CL} \left(\int_0^{\frac{R_a}{R_a + \delta_f}} c_{H_2}(z) z^2 dz \right) x dx$$

★ Where γ, κ, ν, ξ are constants
 $\xi \propto R_a, \delta_f, D_{film}, D_{eff}, S_{film \rightarrow agg}, S_{CL \rightarrow film}$

Transport is sensitive to porosity, particle size

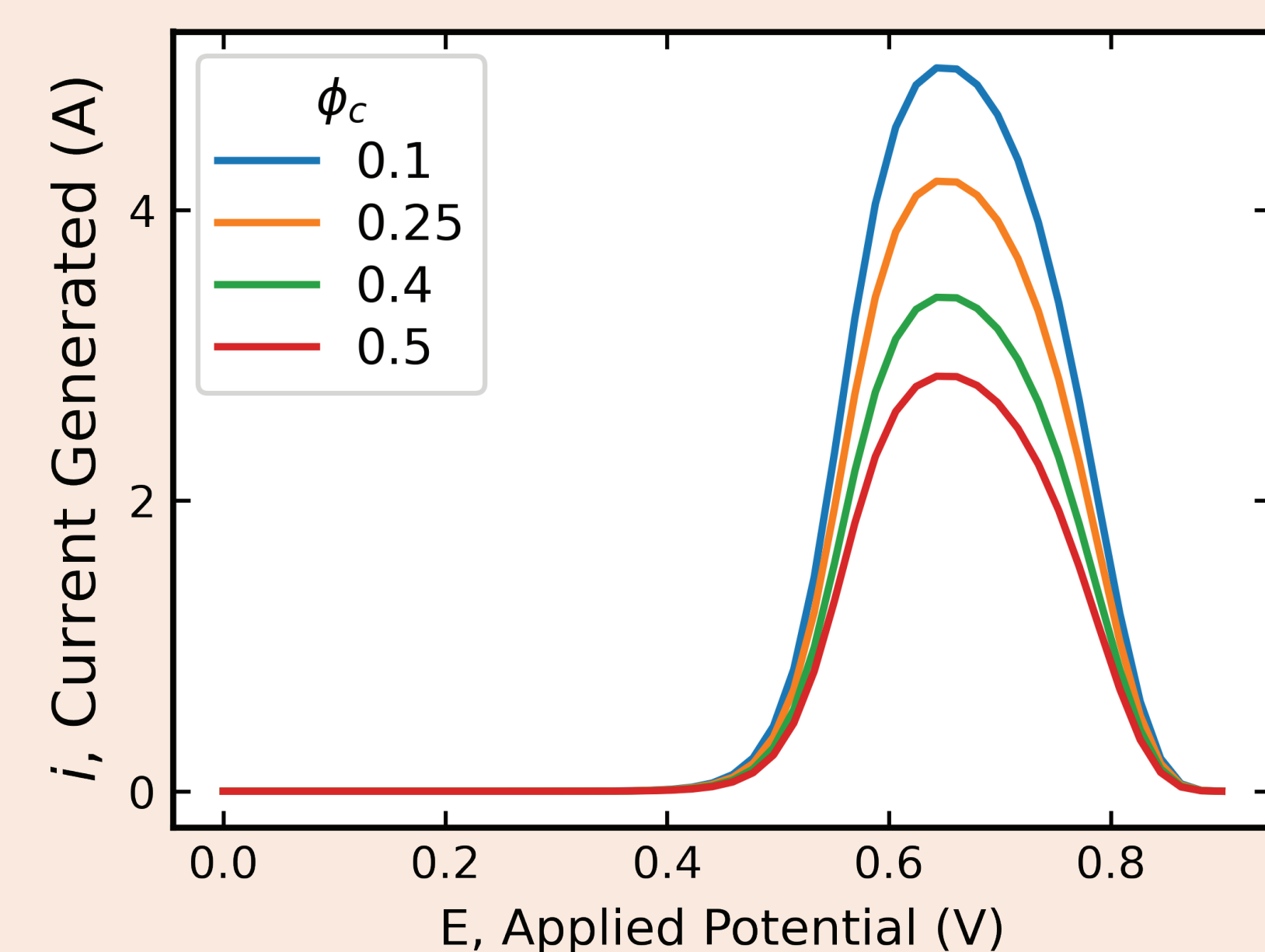


Figure 3. Polarization curve for varying particle porosity.

In **Figure 4** (right):

as $i \rightarrow i_{max}$, $E_R \rightarrow 0$

This behavior is expected; as applied potential increases, reaction kinetics overtake transport and the particle becomes transport-limited.

H₂ concentration within the particle reduces to zero, causing:

$$\nabla c_{H_2} \rightarrow 0$$

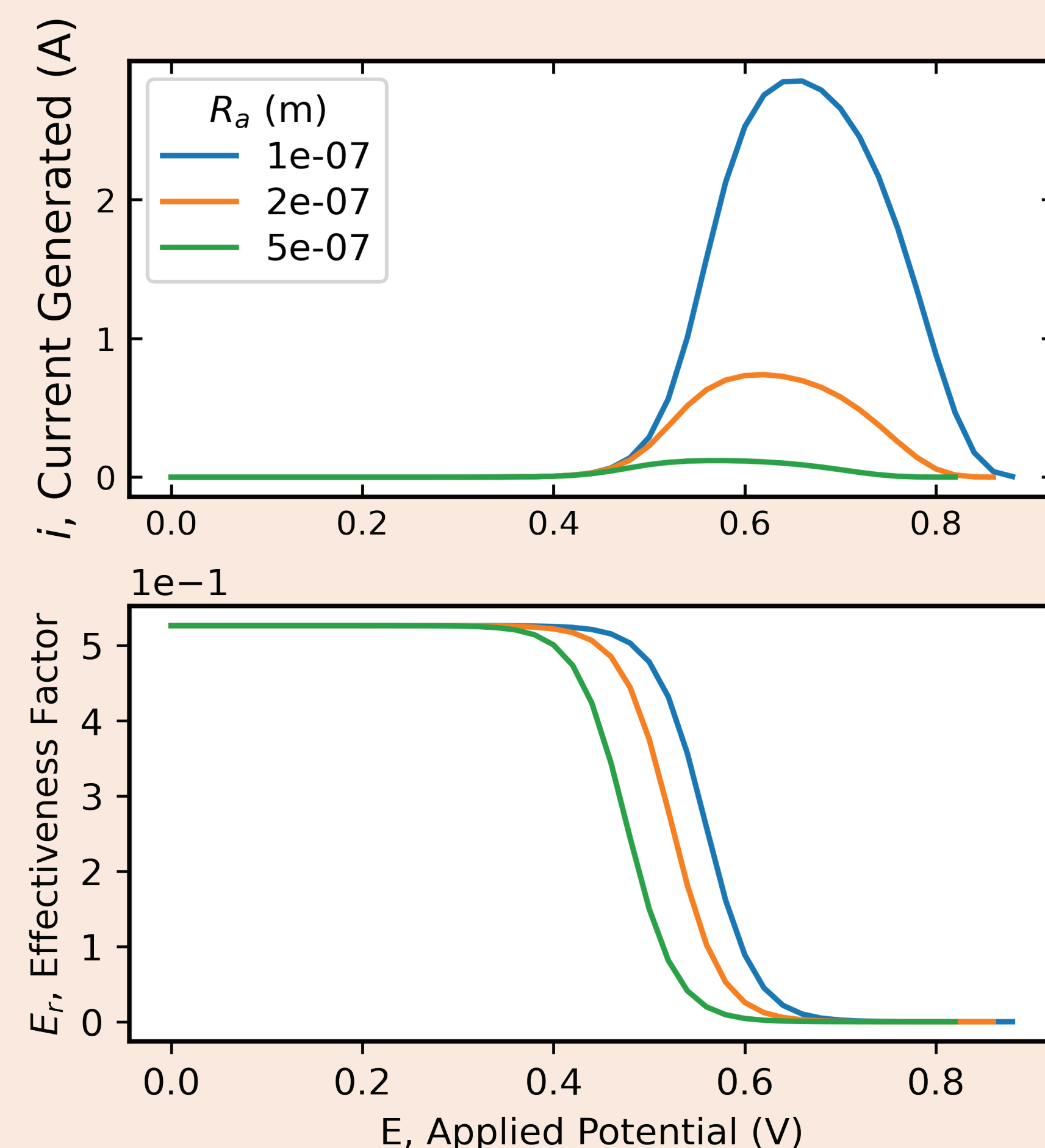
$$W_{AR} \rightarrow 0$$

$$E_R \rightarrow 0$$

Figure 4 (right). Sensitivity analysis of total current generated (top) and effectiveness factors (bottom) for varying catalyst particle radii.

Sensitivity analysis found that **overall, current generation is sensitive to particle size and porosity**. Physically, this is caused by the following:

- Higher porosity reduces available catalyst surface for reaction, slowing down reaction kinetics (**Figure 3**, left).
- Larger particle size measured by radius increases transport resistance, which reduces available H₂.



Catalyst particle is dominantly transport-limited

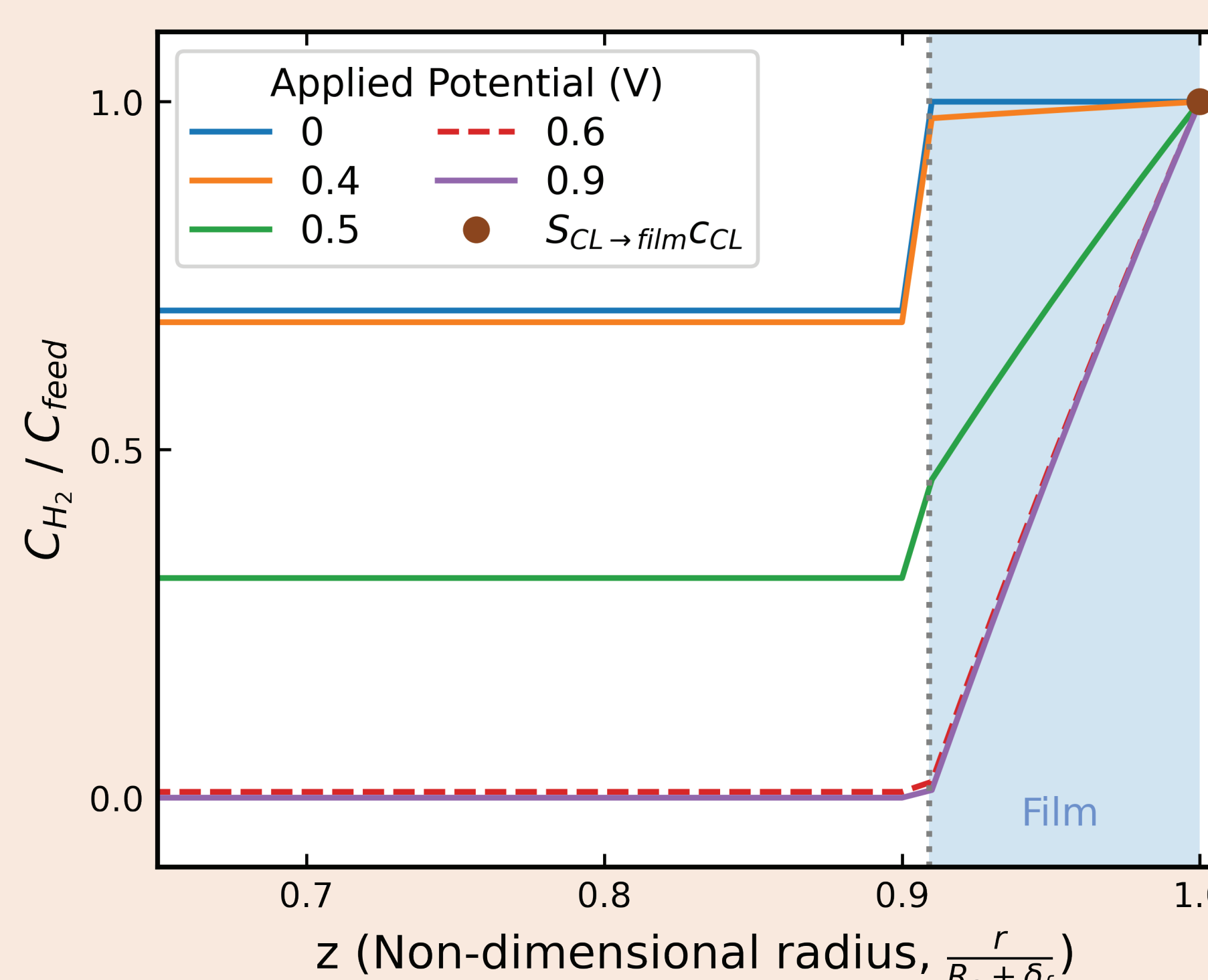


Figure 5. Diagram of H₂ transport through a 1-dimensional radial axis through the catalyst particle film towards the center of the carbon-Pt agglomerate, assumed as a spherical particle.

Notable features of concentration profile across the electrode:

- 1) Steep gradient within the ionomer film layer surrounding the agglomerate.
- 2) At E~1V, the concentration gradient drops to 0 while the CL concentration (not shown) outside of the particle is positive and non-zero.

- 3) H₂ only declines at a small radial volume between $0.9 \geq z \geq z|_{\text{film boundary}}$.

These characteristics imply that:

- 1) The ionomer film poses significant transport resistance H₂ transport into the catalyst particle, especially at high overpotential.
- 2) H₂ transport into the catalyst particle is dominantly transport-limited.
- 3) The catalyst particle is not being fully utilized. Reaction only occurs from the outer 10% of the agglomerate up to the film interface.

Conclusions

This model

- builds upon understanding of H₂ transport resistance within a porous electrode.
- uses purely analytical operations allow for extremely fast simulations of the porous electrode set-up.

Key observations

- Electrode is dominantly transport-limited.
- Electrode reaches an upper-limit of catalyst volume utilization.
- Porosity and size (radius) of the catalyst particle strongly impacts electrode's current generation.

Future development of the model for more sensitive, complex physical behavior to:

- 1) optimizes catalyst particle morphology (radius, catalyst loading, porosity, ionomer content, etc.)
- 2) compare performance enhancement against cost (reducing underutilized catalyst).

Acknowledgements

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